

Total No. of Questions : 8]

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Roll No

EE-7004(3)/EX-7004(4)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

**Renewable and Non-Conventional Energy
Systems**

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) How Renewable energy system differs from non renewable sources?
b) Explain the Energy planning and managements.
2. a) Explain the working principle of wind photovoltaic system with proper diagram. Also write down the disadvantages of using such systems.
b) Explain the operation of wind power generation in an isolated power system.
3. a) Explain solar radiation in brief.
b) Describe the biomass energy systems feeding loads in stand-alone as well as hybrid modes.
4. a) Explain the Solar thermal conversion system.
b) What is the Hybrid system? Explain the working of grid connected hybrid system.

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5.
 - a) Explain power electronic controlled drives for energy saving.
 - b) What are the various measuring devices used for energy measurements?

6.
 - a) Discuss Anderson cycle system for ocean thermal energy conversion and limitations associated with OTEC? Also determine the overall efficiency of an OTEC plant if surface warm water temperature is 27°C and deep cool water temperature is 7°C . Assume relative efficiency factor of the plant 55%.
 - b) What are the energy efficient motors? Discuss about energy saving in power electronic controlled drives?

7.
 - a) What is the methods used for Energy audits?
 - b) How we can save the energy using power electronics drives?

8. Write short notes in account of electricity saving (any two)
 - a) Electricity in pumps
 - b) Air-conditions
 - c) Illumination
